ASSIGNMENT

1. Introduction

This report presents the finite element analysis (FEA) of a humanoid robot leg assembly under realistic loading conditions. The objective is to evaluate the structural performance of the leg when subjected to the full weight of the robot under a worst-case single-leg stance condition. The study includes both a 3D printed anisotropic material model and an isotropic aluminium material model for comparison. The analysis focuses on stress distribution, deformation characteristics, and overall design suitability.

2. Model Understanding

2.1 Assembly Overview

- The assembly consists of nine individual components connected through multiple joint interfaces.
- The upper section contains a light blue clevis, a dark blue pin component, a grey joint component and a black cylindrical connector.
- The middle section consists of a grey structural link which serves as the primary load-transfer member.
- The lower section contains a light blue structural link connected to the central link through a joint interface.
- A yellow cylindrical component forms the lower joint assembly.
- The assembly terminates at a dark blue foot/base support which transfers load to the ground.

2.2 Critical Load-Bearing Components

- Grey central structural link.
- Light blue lower structural link.
- Upper joint assembly.
- Intermediate joint connection.
- Lower joint assembly.
- Foot/base support.

2.3 Potential High-Stress Regions

- Upper pin connection region.
- Black-to-grey transition region.
- Intermediate joint connection between the grey and light blue links.
- Yellow lower joint region.
- Interfaces surrounding pin holes and joint contacts.

3. Assumptions

3.1 Loading Assumptions

- Total robot mass is assumed to be 80 kg as specified in the assignment.
- Gravitational acceleration is taken as 9.81 m/s^2.
- The worst-case loading condition assumes the entire robot weight is supported by a single leg.
- The applied load is calculated as:

  $F = m \times g = 80 \times 9.81 = 784.8\ N \approx 785\ N$

- The load is assumed to act vertically downward through the upper mounting region of the assembly.
- Dynamic effects such as walking, running, jumping, and impact loading are neglected.

3.2 Material Assumptions

- Two material cases are considered for the study.
- Case 1 represents a 3D printed material with anisotropic behaviour.
- Case 2 represents an isotropic aluminium alloy.
- Material properties are assumed uniform throughout each component.
- Material defects and manufacturing imperfections are neglected.

3.3 Joint and Contact Assumptions

- Joint connections are assumed to transfer loads effectively between components.
- Contact surfaces are assumed to be properly assembled without gaps.
- Fasteners, bearings and small hardware details are neglected if they do not significantly influence global structural behaviour.
- Actuators are simplified and modelled as rigid solid bodies as specified in the assignment.

3.4 Analysis Assumptions

- Linear static structural analysis is performed.
- Material behaviour is assumed to remain within the elastic region.
- Large deformation effects are neglected.
- Thermal effects are not considered.
- The assembly is assumed to be free from initial damage or manufacturing defects.

4. Design & Manufacturing Considerations

4.1 CAD to STL Preparation

- The CAD model will be converted into STL format before 3D printing.

- **A fine STL resolution** will be used to maintain the shape and dimensions of the components.
- Important features such as pin holes, joints and load-bearing regions will be preserved.
- Very coarse STL settings will be avoided to prevent loss of accuracy.

4.2 Part Orientation Strategy

- Parts will be oriented to achieve maximum strength during printing.
- The print direction will be selected based on the expected load on each component.
- **The weaker layer direction of the printed material** will be considered.
- Support material will be minimized wherever possible.
- A balance between strength, print quality and printing time will be maintained.

4.3 Assembly Strategy

- The leg assembly is recommended to be printed as **separate components.**
- Printing individual parts allows better print orientation for each component.
- Separate parts are easier to manufacture and assemble.
- Damaged parts can be replaced without reprinting the entire assembly.
- The existing pins and joints can be used to assemble the final structure.

5. Material Selection

5.1 3D Printed Material (PETG)

- PETG is selected as the material for the 3D printed version of the leg.
- PETG is commonly used for strong and functional 3D printed parts.
- The material is assumed to be anisotropic because its properties depend on the print direction.
- The strength of the printed part is lower in the layer bonding direction compared to the layer plane.
- The structural links are assumed to be printed such that the primary load acts along the stronger layer direction.
- Exact directional material data is not available, so standard material properties are used.
- The effect of print orientation is considered while evaluating the results.

Material Properties (PETG)

Property	Value
Density	1270 kg/m³
Young's Modulus	2.1 GPa

Property	Value
Poisson's Ratio	0.38

5.2 Aluminium Material (6061-T6 Aluminium)

- 6061-T6 Aluminium is selected for the comparison study.
- The material is assumed to be isotropic, meaning its properties are the same in all directions.
- Aluminium provides higher stiffness and strength compared to PETG.
- The material is widely used in structural and robotic applications.
- The same geometry, loading conditions and boundary conditions are used for both material cases to enable a fair comparison.

Material Properties (6061-T6 Aluminium)

Property	Value
Density	2700 kg/m³
Young's Modulus	69 GPa
Poisson's Ratio	0.33
Yield Strength	276 MPa

3. Boundary Conditions & Loading

To simulate the structural behaviour of the robot leg assembly under static loading conditions, appropriate boundary conditions and external loads were applied.

Ground Constraint

The bottom surface of the foot support, representing the contact region between the robot leg and the ground, was assigned a fixed boundary condition. This constraint restricts all translational and rotational degrees of freedom and provides a stable reference for load transfer through the assembly.

Applied Load

A downward static load of 785 N was applied at the upper clevis region of the assembly. The load represents the body weight transferred from the robot chassis to the leg during operation.

Load Transfer Assumption

The load was assumed to be transferred through all structural members and joint interfaces of the leg assembly. Tie constraints were used between mating components to ensure continuous load transfer throughout the assembly. This assumption represents a bonded connection between adjacent components and enables evaluation of the overall structural response under static loading conditions**.**

4. Meshing

Finite Element Analysis (FEA) requires discretization of the geometry into smaller elements to accurately predict stress and deformation behaviour.

<u>Element Selection</u>

Three-dimensional solid elements were used to represent the structural components of the robot leg assembly. These elements are suitable for capturing the complex geometry and stress distribution within the assembly.

<u>Mesh Generation</u>

A global mesh was generated for the complete assembly using the meshing tools available in PrePoMax. The mesh density was selected to achieve a balance between computational efficiency and solution accuracy.

<u>Local Refinement</u>

Special attention was given to regions expected to experience high stress concentrations, including:

- Upper clevis connection region
- Pin-hole interfaces
- Intermediate joint connection
- Lower joint assembly
- Foot support connection region

A finer mesh was adopted around these locations to improve the accuracy of stress prediction and deformation results.

<u>Mesh Strategy Justification</u>

The selected meshing strategy was based on the expected load path through the assembly. Regions with geometric discontinuities and joint interfaces were meshed more finely due to their susceptibility to stress concentration effects. Less critical regions were meshed with comparatively larger elements to reduce computational time while maintaining acceptable solution quality.

5. Analysis and Interpretation

5.1 Grey Central Structural Link

Results:

Aluminium 6061-T6

Maximum Deformation = 4.656 mm

Maximum Von Mises Stress = 251 MPa

PETG

Maximum Deformation = 151.9 mm

Maximum Von Mises Stress = 248.4 MPa

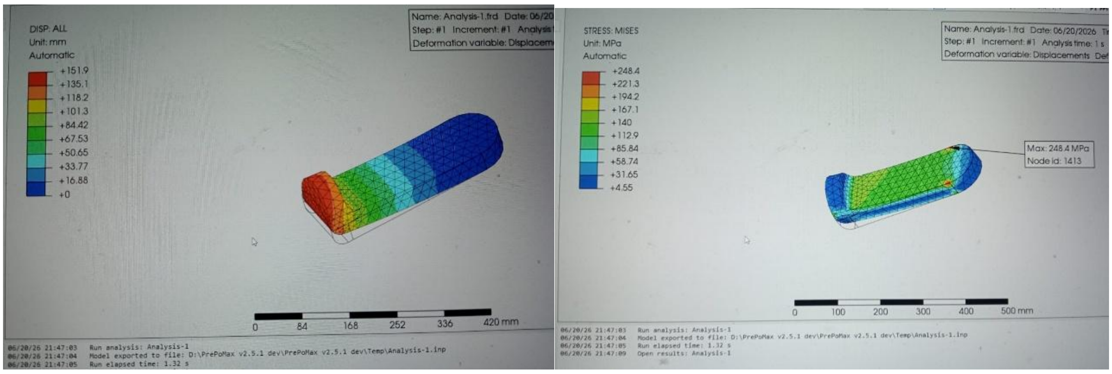
Interpretation

Maximum deformation was observed at the upper circular connection region for both materials. The maximum Von Mises stress occurred near the lower connection interface, indicating stress concentration in the load transfer path.

PETG exhibited significantly higher deformation compared to Aluminium 6061-T6 due to its lower stiffness. However, both materials showed similar stress concentration locations, indicating that the geometry and load path primarily govern the stress distribution.

The lower connection interface was identified as the most critical area of the component from a structural integrity perspective.

PETG:



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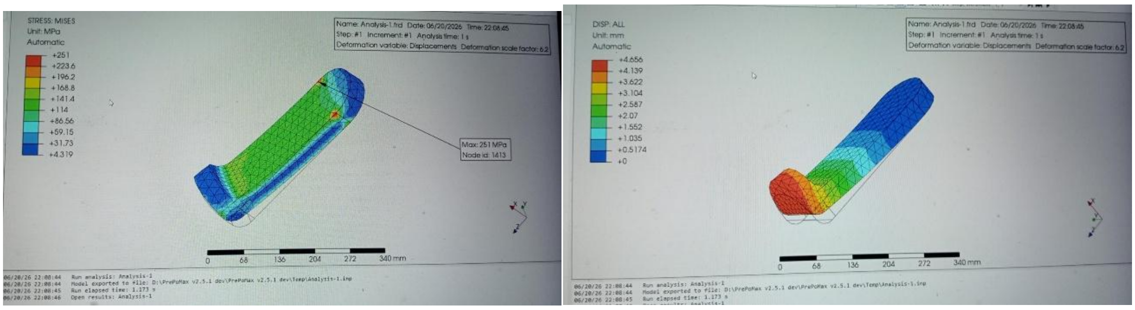


5.2 Light Blue Lower Structural Link

Results

Aluminium 6061-T6

Maximum Deformation = 0.08567 mm

Maximum Von Mises Stress = 28.42 MPa

PETG

Maximum Deformation = 2.809 mm

Maximum Von Mises Stress = 28.25 MPa

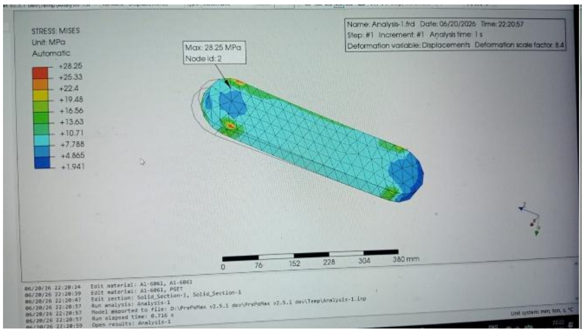
Interpretation

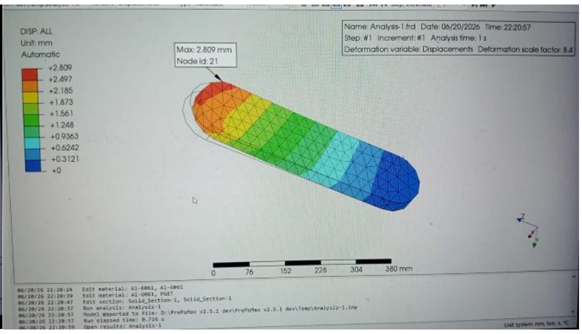
Maximum deformation was observed at the upper region of the component for both materials. The maximum Von Mises stress occurred at localized regions near the upper connection interface, indicating stress concentration at the load application area.

PETG exhibited significantly higher deformation compared to Aluminium 6061-T6 due to its lower stiffness. However, both materials showed nearly identical stress levels and similar stress concentration locations, indicating that the geometry and load path primarily govern the stress distribution.

The upper connection region was identified as the most critical area of the component from a structural integrity perspective.

PETG:





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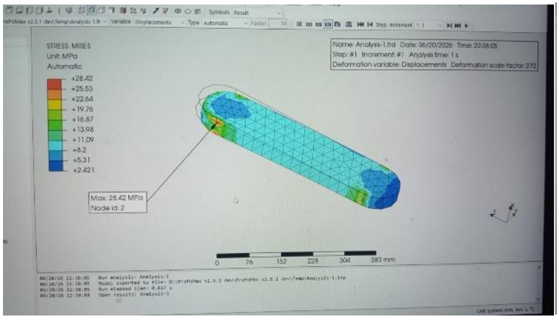
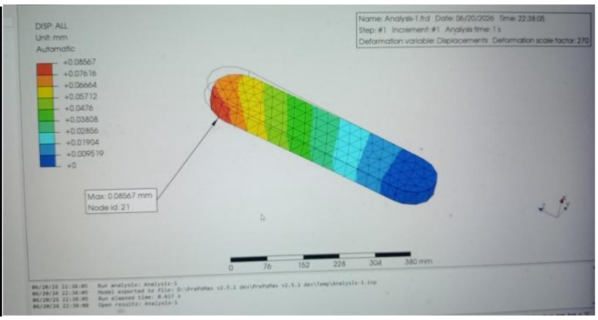




5.3 Upper Joint Assembly

Results

Aluminium 6061-T6

Maximum Deformation = 5.241 mm

Maximum Von Mises Stress = 1762 MPa

PETG

Maximum Deformation = 167.6 mm

Maximum Von Mises Stress = 1760 MPa

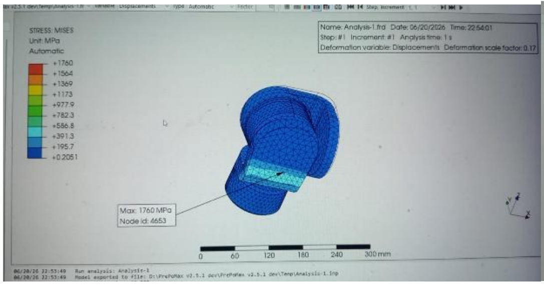
Interpretation

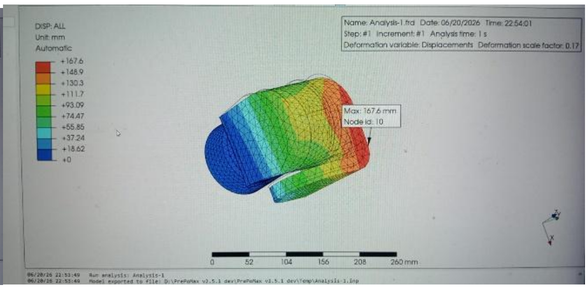
Maximum deformation was observed at the upper clevis region for both materials. The maximum Von Mises stress occurred at a localized region on the inner surface of the grey component near its connection with the blue connector, indicating stress concentration at the joint interface.

PETG exhibited significantly higher deformation compared to Aluminium 6061-T6 due to its lower stiffness. However, both materials showed nearly identical stress levels and similar stress concentration locations, indicating that the geometry and load path primarily govern the stress distribution.

The connection region between the grey component and the blue connector was identified as the most critical area of the assembly from a structural integrity perspective.

PETG:





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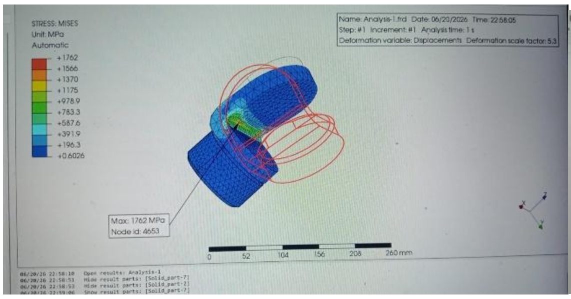
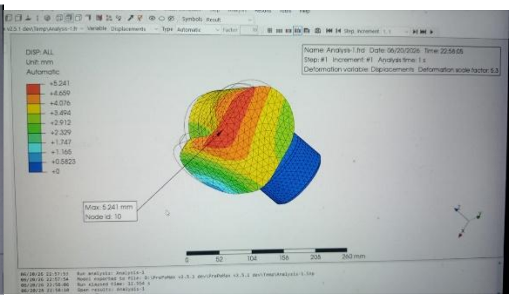




5.4 Intermediate Joint Connector

Results

Aluminium 6061-T6

Maximum Deformation = 0.1671 mm

Maximum Von Mises Stress = 420.6 MPa

PETG

Maximum Deformation = 5.546 mm

Maximum Von Mises Stress = 409.3 MPa

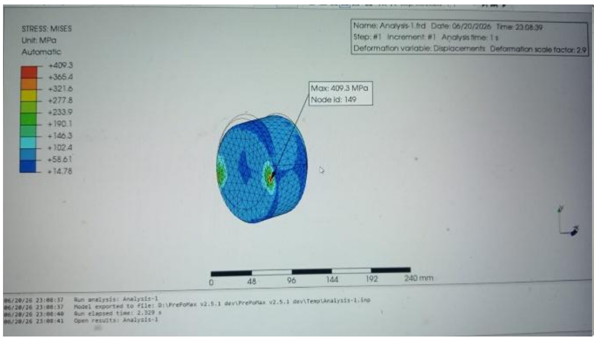
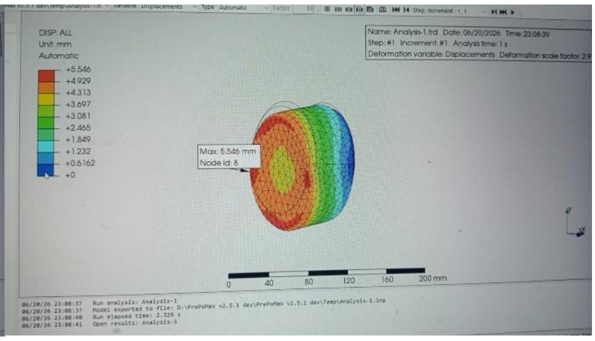
Interpretation

Maximum deformation was observed at the cylindrical surface connected to the grey link for both materials. The maximum Von Mises stress occurred at localized regions near the connector interfaces, indicating stress concentration in the load transfer path.

PETG exhibited significantly higher deformation compared to Aluminium 6061-T6 due to its lower stiffness. However, both materials showed similar stress levels and stress concentration locations, indicating that the geometry and load path primarily govern the stress distribution.

The connector interfaces between the grey link and blue link were identified as the most critical regions of the component from a structural integrity perspective.

PETG:

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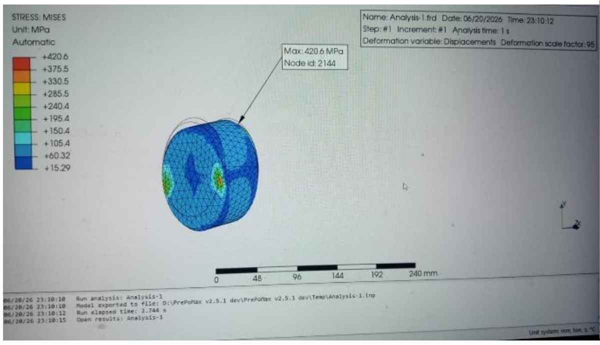
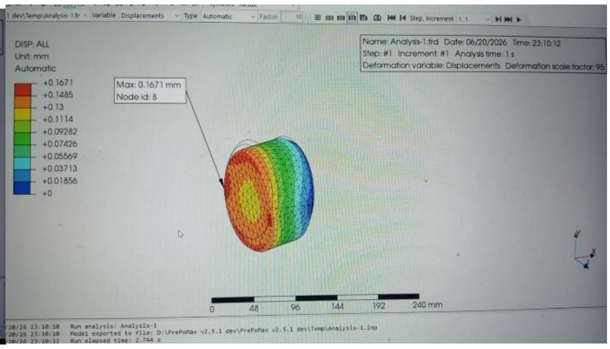



5.5 Foot Support

Results

Aluminium 6061-T6

Maximum Deformation = 2.399 mm

Maximum Von Mises Stress = 256.1 MPa

PETG

Maximum Deformation = 77.41 mm

Maximum Von Mises Stress = 253.7 MPa

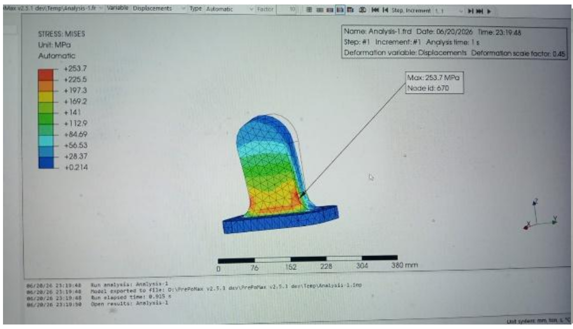
Interpretation

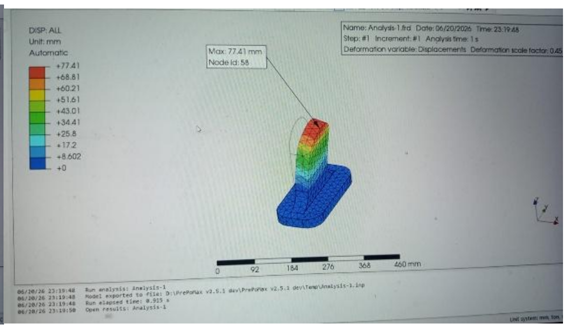
Maximum deformation was observed at the curved upper region of the foot support for both materials. The maximum Von Mises stress occurred at localized regions along the edge connecting the vertical support section and the flat base section, indicating stress concentration at the geometric transition.

PETG exhibited significantly higher deformation compared to Aluminium 6061-T6 due to its lower stiffness. However, both materials showed similar stress levels and stress concentration locations, indicating that the geometry and load path primarily govern the stress distribution.

The junction between the vertical support and the flat base was identified as the most critical region of the component from a structural integrity perspective.

PETG:





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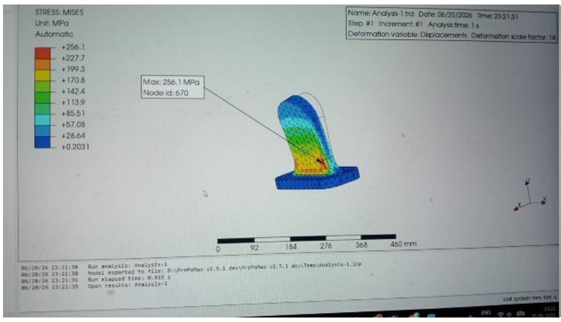
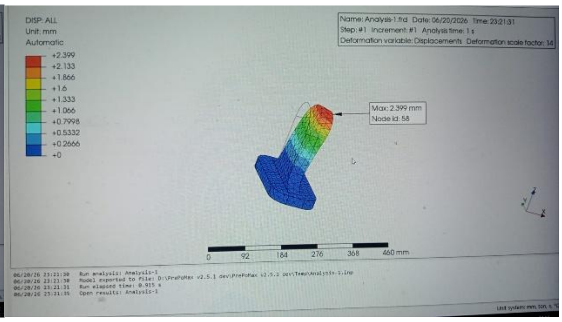




5.6 Complete Assembly Analysis

Results

Aluminium 6061-T6

Maximum Deformation = 367 mm

Maximum Von Mises Stress = 6338 MPa

PETG

Maximum Deformation = 1.196×10^4 mm

Maximum Von Mises Stress = 5980 MPa

Interpretation

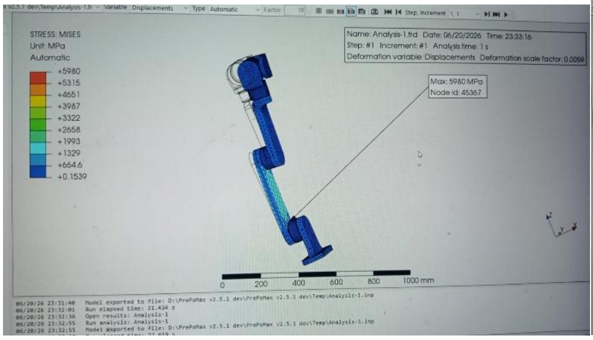
The highest stress concentrations were observed at the interface between the blue lower link and the lower connector connecting the assembly to the foot support.

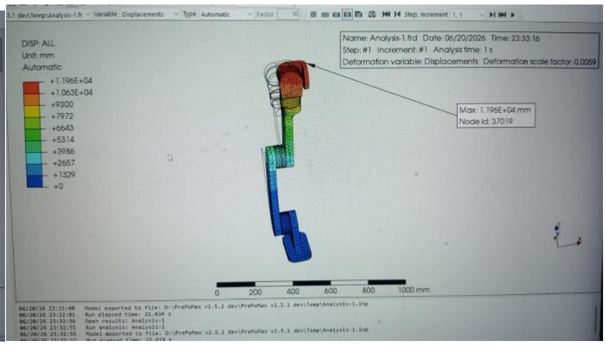
From a structural safety perspective, the assembly exhibits significant deformation and high localized stress values under the applied loading conditions. These results indicate that the joint regions play a dominant role in determining the structural response of the assembly and should be considered carefully during future design optimization.

Significant stress concentration effects were observed at the connection interfaces and geometric discontinuities where load transfer occurs between components. Maximum deformation was observed at the upper clevis region, with displacement increasing progressively from the fixed support towards the free end of the assembly, indicating bending-dominated structural behaviour.

Several modelling limitations should be considered while interpreting the results. The imported CAD assembly did not contain mate definitions, revolute joints, or kinematic constraints that would normally represent the actual behaviour of the robot leg. Tie constraints were therefore used to ensure load transfer between components, causing the assembly to behave as a bonded structure rather than a true pin-jointed mechanism. In addition, the peak stress values were observed over very small localized regions near joint interfaces, which may be influenced by stress concentration effects and numerical stress singularities. Consequently, the results should be interpreted as representative of the adopted modelling assumptions rather than exact real-world behaviour.

PETG:





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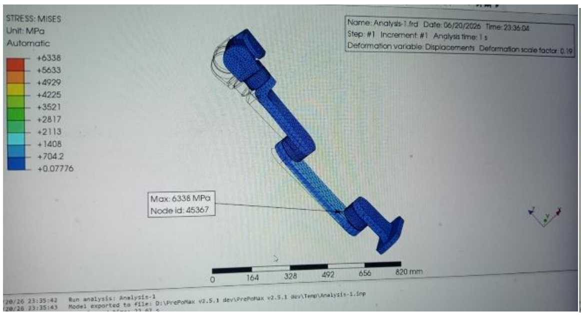

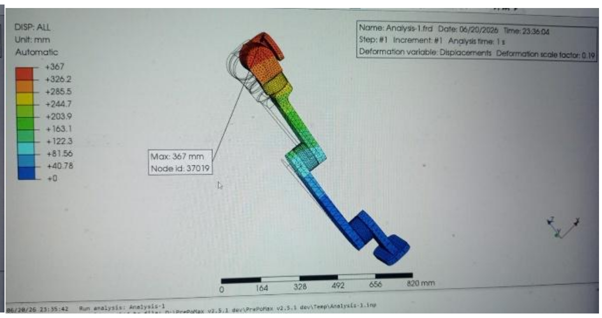

7.Design Suggestions

Based on the results obtained from the finite element analysis, the following design improvements are recommended:

1. Strengthening of the Lower Joint Connection

The highest stress concentrations in the complete assembly were observed at the interface between the blue lower link and the lower connector attached to the foot support. This region should be reinforced by increasing the local cross-sectional area, adding fillets at sharp transitions, or redesigning the connection geometry to reduce stress concentration and improve structural reliability.

2. Improvement of Upper Clevis Stiffness

The maximum deformation in the assembly occurred at the upper clevis region for both PETG and Aluminium 6061-T6. Increasing the stiffness of this region through local reinforcement, increased section thickness, or improved geometric support can reduce overall assembly deformation and improve structural stability.

3. Optimization of Joint Interfaces

Several component analyses indicated that the highest stress values occurred at connection interfaces and geometric transitions. Smoother load transfer between connected components can be achieved by increasing fillet radii and reducing abrupt geometric changes at these locations, thereby minimizing stress concentration effects.

4. Material Selection Consideration

The analysis showed that PETG experienced significantly higher deformation than Aluminium 6061-T6 under identical loading conditions. For applications requiring high stiffness and dimensional stability, Aluminium 6061-T6 is the preferred material. PETG may still be suitable for lightweight prototypes and low-load applications where large deformation can be tolerated.